

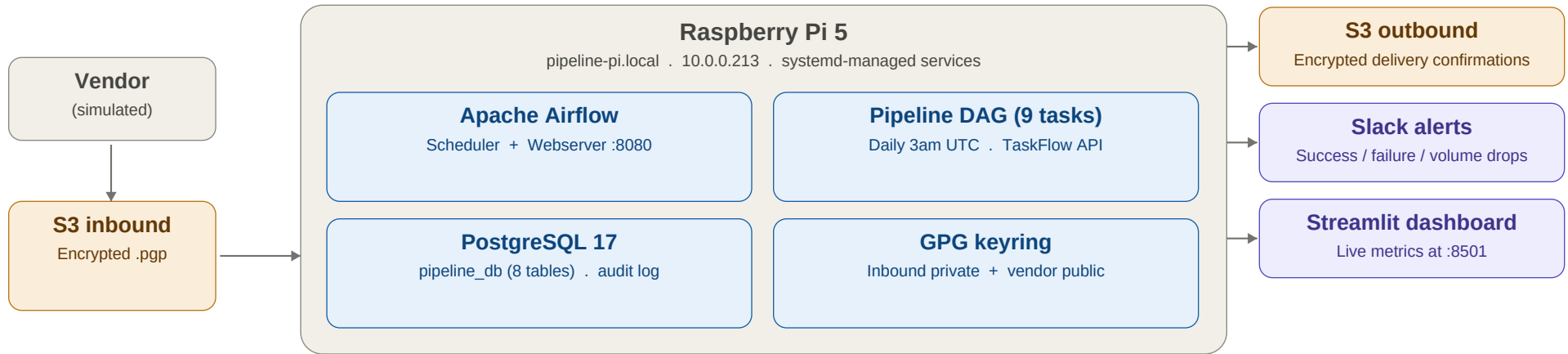
# Northstar Encrypted Data Pipeline

End-to-end PGP-encrypted . AWS S3-integrated . Apache Airflow . Raspberry Pi 5

[github.com/grant833/encrypted-data-pipeline](https://github.com/grant833/encrypted-data-pipeline)

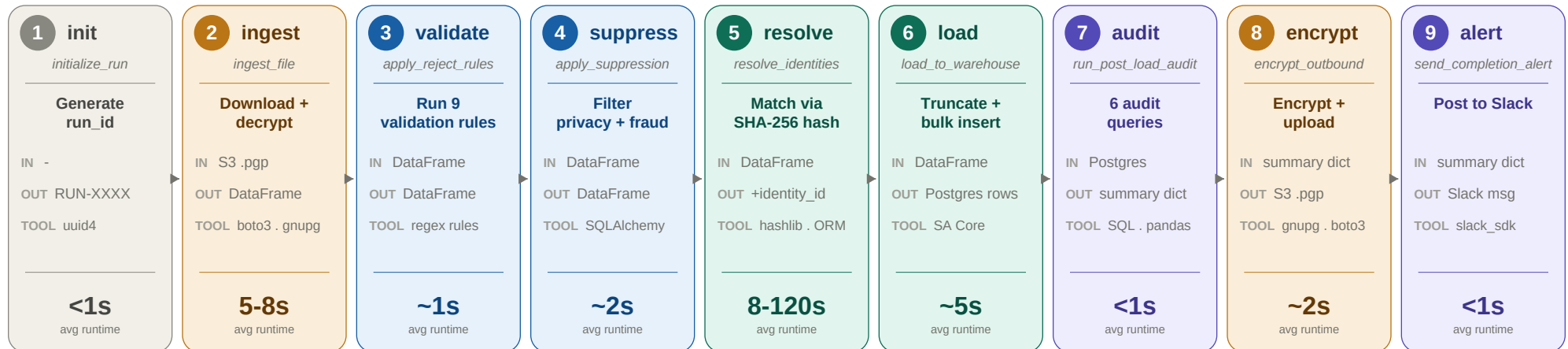
9 tasks . 50K records / run . daily 3am UTC

## 1. System architecture



Detailed pipeline flow inside the DAG

## 2. Pipeline DAG breakdown . customer\_application\_pipeline



**Stack** Python 3.13 . PostgreSQL 17 . Apache Airflow 2.11 . GPG/PGP . AWS S3 (boto3) . Slack SDK . Streamlit . Plotly . pandas . SQLAlchemy . pytest . systemd

**Tables (8)** customer\_applications . identities . pipeline\_audit\_log . privacy\_opt\_outs . privacy\_deletes . fraud\_watchlist . suppression\_exclusions . rejected\_records